

Telecom Portfolio Risk Report

Generated 17 August 2026 · portfolio state, not a time series

Unmarked	OBSERVED - measured from the source data.
†	PROJECTED - derived from the stated economic assumptions, not measured.
‡	SIMULATED - from generated history. Shows portfolio structure only; churn in it is flat by construction and cannot support a forecast.

Portfolio at a glance

3,333	14.49%	483
Customers	Churn rate	Churned
\$198,146	\$59.45	\$31,567
Revenue	ARPU	Revenue at risk

The portfolio is seeing a churn rate of 14.49%, leaving 31 566.93 of revenue at risk. Churn is driven by three independent factors: customers with 4+ service calls (267 customers), international plan subscribers (323 customers, 9.69% of the base), and heavy daytime usage (210 customers). The service call cliff is the most immediate risk because churn jumps sharply once customers exceed three calls. Consequently, the high service call segment should be addressed first with proactive, fast track support to reduce churn. Implement a targeted outreach program for these 267 customers to protect the at risk revenue.

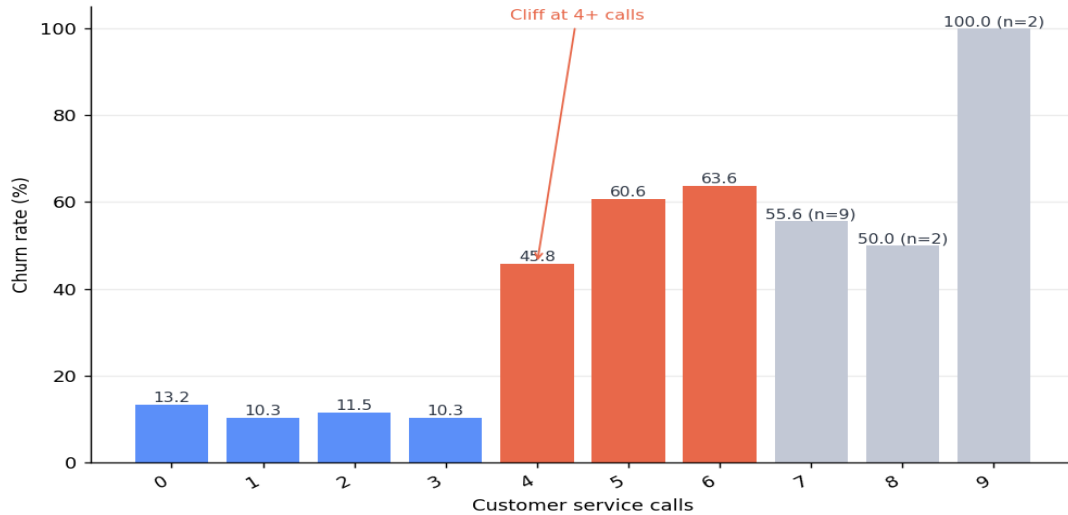
Confirmed churn drivers

First, customers who have made 4 or more service calls churn at 51.69 % versus 11.25 % for those with 0-3 calls; the threshold is four calls, and because churn jumps sharply at that point we should intervene proactively once a customer reaches the fourth call. Second, having an international plan is a threshold driver: customers without an international plan churn at 13.85 % (no voicemail) or 5.3 % (voicemail), while those with an international plan churn at 43.72 % (no voicemail) or 39.13 % (voicemail); we should focus retention offers on the international plan segment. Third, heavy daytime usage (day_charge ≥ 45) shows churn of 60.0 % versus 11.43 % for normal usage; the threshold is a day charge of 45, so high usage accounts should be flagged for targeted retention actions.

Driver	Threshold	Below	Above	Customers
Customer service calls	4 or more	11.25%	51.69%	267
International plan	subscribed	11.5%	42.41%	323
Daytime charge	\$45 or more	11.43%	60.0%	210

Churn by number of customer service calls

Churn is flat from 0 to 3 calls, then jumps sharply. This is a cliff, not a slope.

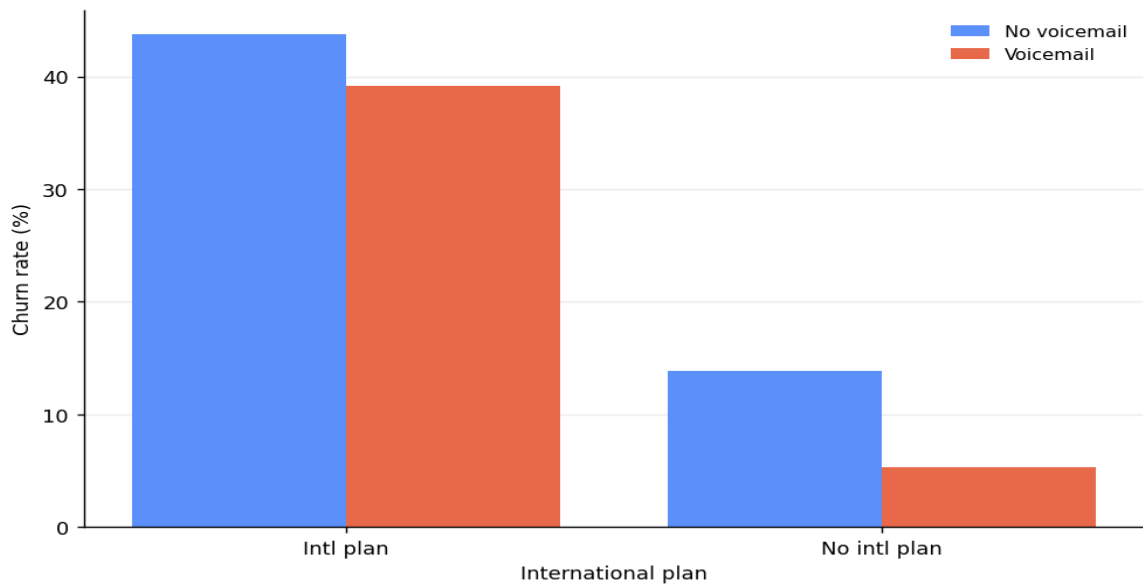


Bucket as '0-3' vs '4+'. Treating call count as a continuous predictor understates the effect badly. Greyed bars have fewer than 20 customers and are too small to read as evidence.

AI explanation. The chart shows churn rates staying between 13.2% and 10.33% for customers with 0 to 3 service calls, then jumping sharply to 45.78% at the fourth call; rates stay above 60% for the next two call levels where the sample size remains adequate. This cliff suggests that reaching a fourth call may be a proxy for unresolved frustration or that the customer has already decided to leave before calling again. Implement an automatic trigger to flag any customer after their third call and assign a senior support specialist to resolve the issue proactively, aiming to prevent the fourth call escalation.

Churn by plan combination

International plan holders churn several times higher.

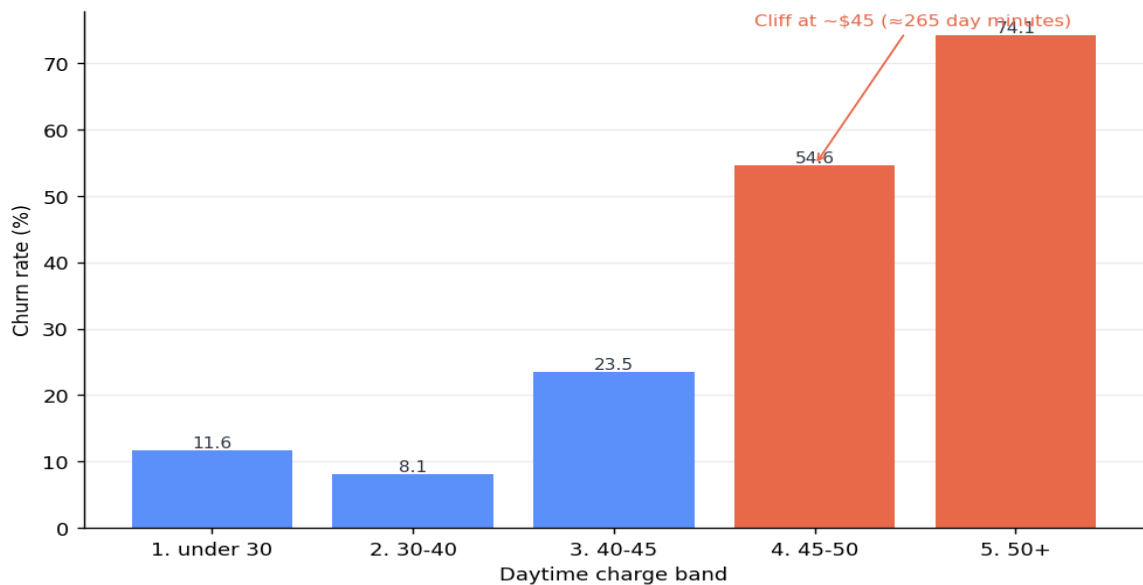


Voicemail plan has little independent effect; the international plan is the driver.

AI explanation (UNVALIDATED). The chart shows two distinct clusters: customers with an international plan have churn rates around 40 % (43.72 % without voicemail, 39.13 % with voicemail), while those without an international plan sit much lower at 13.85 % (no voicemail) and 5.3 % (voicemail). The shift between the “Intl plan” and “No intl plan” groups is sharp, whereas adding voicemail moves the churn rate only a few points within each international plan segment. This pattern suggests that the international offering itself is the primary driver of churn, perhaps because its price-value balance or fit is unattractive to users, while voicemail status has little impact. A concrete step is to pilot a revised international plan pricing or feature bundle for the high churn segment and measure whether churn drops relative to the current 40 % level.

Churn by daytime charge band

Churn DIPS before it explodes. A linear model of day charge would find almost nothing.



Found while building the Phase 6 rules engine. The earlier two-driver segmentation reported a 'baseline' of 8.2% that concealed this group.

AI explanation. The churn rate falls from 11.64 % in the “under 30” band to 8.12 % in the “30-40” band, then jumps to 23.53 % in the “40-45” band and climbs sharply to 54.61 % and 74.14 % in the two highest bands. This dip-then-spike shape suggests that modest daytime spenders are relatively comfortable, while customers whose daytime usage becomes heavy enough to push them into higher charge brackets may be hitting bill shock or a tariff that does not fit their usage pattern. The steep rise aligns with average daytime minutes rising from about 134.5 to over 276 minutes as the charge bands increase. Launch a pilot offering a capped or discounted plan for customers in the 45-50 and 50+ bands to test whether a better-matched tariff reduces churn.

Risk segments and recommended actions

****1 – Critical – all three risk drivers (R1_triple)**** – targets the single customer who shows 4 + service calls, an international plan and heavy daytime usage. Make an immediate senior level retention call the same day and offer premium support, a tariff review and a 20 % discount for three periods. The data shows this customer already churned (100 % churn rate) and represents £81.98 of revenue at risk; the projected net value of the intervention is £241.6 with an ROI of 360.5 % (assumptions based, not measured). ****2 – Severe – 4 + service calls plus a second driver (R2_calls_plus)**** – targets the 42 customers who have 4 + calls and either an international plan or heavy daytime usage. Deploy a senior retention specialist call within 48 hours and provide premium support plus a 15 % discount for three periods. These customers have a churn rate of 66.67 % and £1,834.15 of revenue at risk; the projected net value is £5,752.2 with an ROI of 428.1 % (based on assumptions). ****3 – Severe – heavy daytime usage (R3_heavy_day)**** – targets the 194 customers whose day charge is ≥ 45. Arrange a tariff review call within five working days and propose moving them to a daytime bundle or higher allowance plan. Their churn rate is 59.28 % and they hold £9,225.97 of revenue at risk; the projected net value is £29,016.57 with an ROI of 607.5 %

(assumption-driven). **4 – High – 4 + service calls only (R4_high_calls)** – targets the 224 customers with 4 + calls but no other driver. Make a proactive support review call within five working days and offer a service credit plus a dedicated support contact. These customers churn at 48.66 % and represent £5,548.55 of revenue at risk; the projected net value is £20,044.0 with an ROI of 494.3 % (projected). **5 – Elevated – international plan only (R5_intl_plan)** – targets the 267 customers on an international plan. Conduct a plan review conversation and offer tariff optimisation or a penalty-free downgrade. Their churn rate is 37.83 % with £6,138.72 of revenue at risk; the projected net value is £20,823.26 and ROI 974.9 % (derived from assumptions). *All projected net values and ROI percentages rely on the stated economic assumptions and are not direct measurements from the snapshot.*

Segment	Customers	Churn	Revenue at risk	Net value†
Critical — all three risk drivers	1	100.0%	\$82	\$242†
Severe — 4+ service calls with a second driver	42	66.67%	\$1,834	\$5,752†
Severe — heavy daytime usage	194	59.28%	\$9,226	\$29,017†
High — 4+ service calls only	224	48.66%	\$5,549	\$20,044†
Elevated — international plan only	267	37.83%	\$6,139	\$20,823†
Baseline — no risk drivers	2,605	4.95%	\$8,738	-

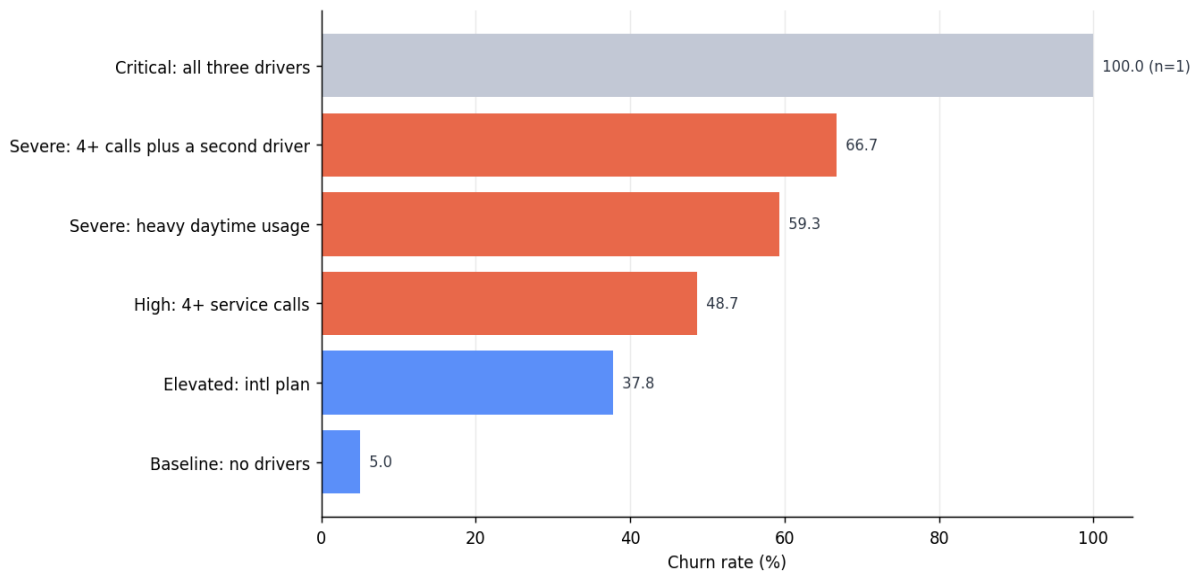
Recommended actions

Segment	Action	Offer	Customers
Critical — all three risk drivers	Immediate senior retention call, same day	Premium support + tariff review + 20% discount, 3 periods	1
Severe — 4+ service calls with a second driver	Senior retention specialist call within 48 hours	Premium support tier + 15% discount for 3 periods	42
Severe — heavy daytime usage	Tariff review call within 5 working days	Move to a daytime bundle or higher-allowance plan	194
High — 4+ service calls only	Proactive support review call within 5 working days	Service credit + dedicated support contact	224
Elevated — international plan only	Plan review — check the tariff fits actual usage	Tariff optimisation or downgrade without penalty	267

† **Assumptions behind projected values:** save rate 30% (band 15%-45%), contact cost \$8.00, acquisition cost \$45.00, horizon 12 periods. These are industry-typical placeholders, not measurements: the dataset contains no cost data.

Churn rate by risk segment

Segments built from the three confirmed drivers.

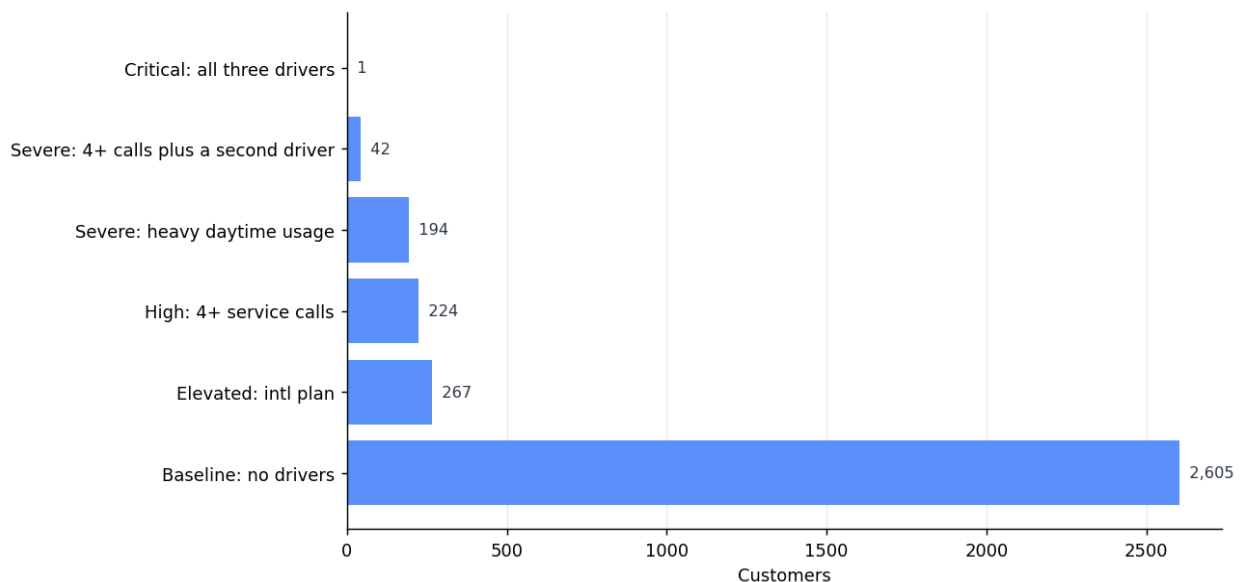


Segments are mutually exclusive and exhaustive: counts sum to 3,333 and revenue to the portfolio total. Greyed bars have fewer than 20 customers and are too small to read as evidence.

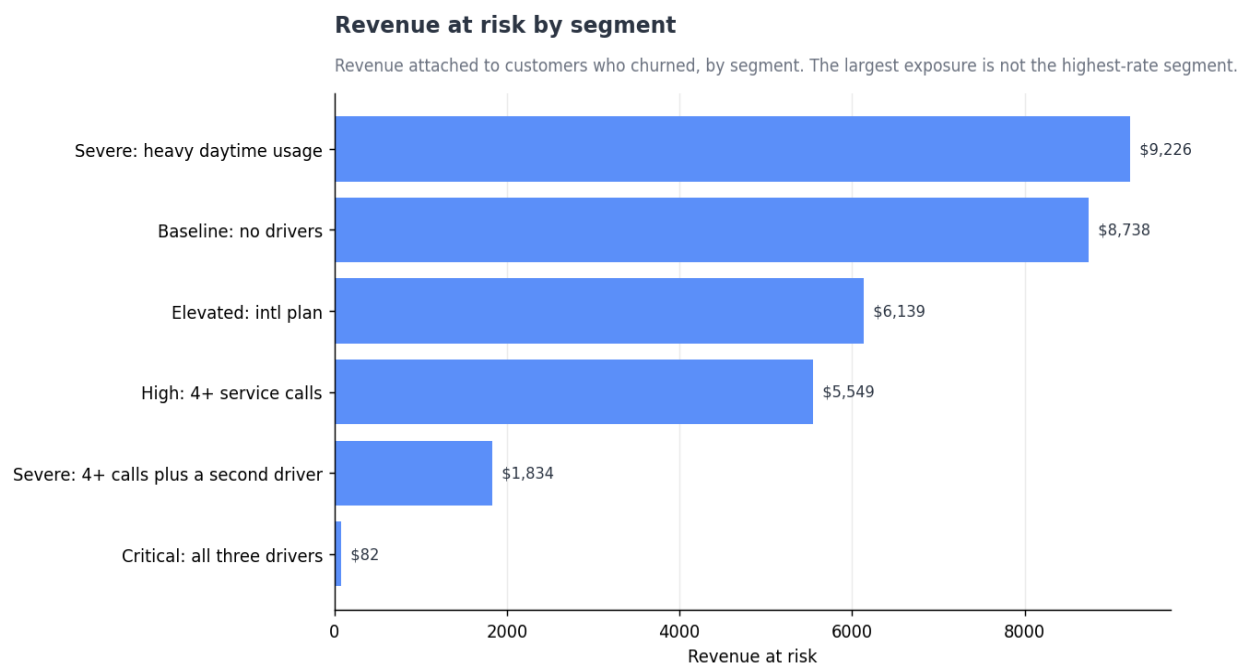
AI explanation. The churn rate falls sharply from 66.67 % in the “Severe: 4+ calls plus a second driver” segment to 59.28 % in “Severe: heavy daytime usage,” then declines more gradually through the “High: 4+ service calls,” “Elevated: intl plan,” and finally the “Baseline” segment at 4.95 %. This pattern suggests that customers exposed to multiple risk factors—especially the combination of frequent service calls and another driver—are far more likely to leave than those with a single driver. Because the “Severe: 4+ calls plus a second driver” group has both a high churn rate and a sizable base of 42 customers (representing \$1,834.15 revenue at risk), it should be tackled first. Deploy a dedicated retention campaign to those 42 customers, offering proactive support and a customized usage review incentive to address both call volume and the secondary risk factor.

Customers per risk segment

Most of the base carries no risk driver at all.



Segment sizes are highly uneven; read rates alongside n.



Rate and exposure rank differently. A large low-rate segment can carry more absolute revenue risk than a small high-rate one.

AI explanation. The chart shows a steep peak at the “Severe: heavy daytime usage” segment with \$9,225.97 revenue at risk, then a sharp decline to the “Baseline: no drivers” segment at \$8,737.56 and further drops to \$6,138.72, \$5,548.55, \$1,834.15, and \$81.98 for the remaining groups. Although the heavy daytime segment’s churn rate is 59.28 percent—lower than the 66.67 percent in the “Severe: 4+ calls plus a second driver” segment and far below the 100 percent in the single customer “Critical” segment—it carries the largest absolute dollar exposure because it combines a sizable customer base (194 customers) with moderate churn. This pattern suggests that a large, moderately risky cohort can generate more revenue loss than a tiny, very high-risk cohort. To protect the most revenue, the business should design a focused retention program for the heavy daytime usage customers, such as targeted usage-based incentives or proactive service outreach.

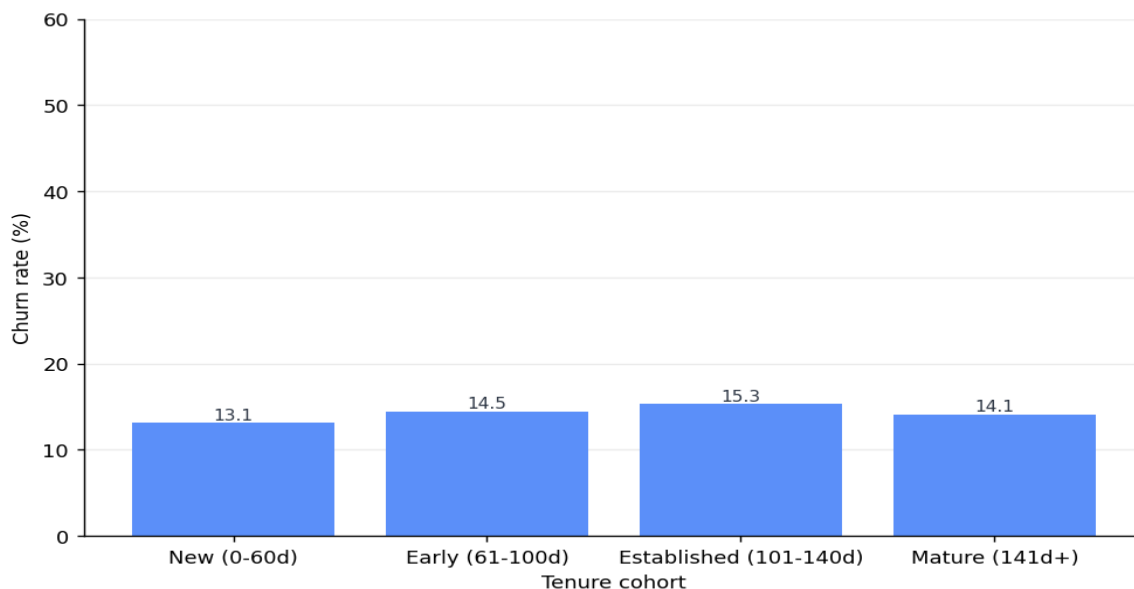
Tenure: an expected driver that is not one

The churn rates for the four tenure cohorts—13.11 %, 14.48 %, 15.29 % and 14.12 %—are all within a narrow band, showing that length of relationship does not separate churn risk. This tells us that tenure is not a useful lever for retention; instead, the data highlight behavioural drivers such as having 4 or more service calls, which pushes churn to roughly 50 %–53 % across every cohort. Because those behavioural signals, not tenure, drive loss, the retention program should focus on customers with 4+ service calls (and also on international plan subscribers and heavy daytime users) rather than on how long they have been with the company. Targeted outreach, proactive support and usage-based incentives for those high-risk behaviours will be far more effective than tenure-based segmentation.

Cohort	Customers	Churn	ARPU	Avg service calls
New (0-60d)	511	13.11%	\$59.70	1.52
Early (61-100d)	1,153	14.48%	\$59.13	1.57
Established (101-140d)	1,138	15.29%	\$59.49	1.6
Mature (141d+)	531	14.12%	\$59.83	1.5

Portfolio structure by tenure cohort

Churn is flat across cohorts: 13.1% to 15.3%.

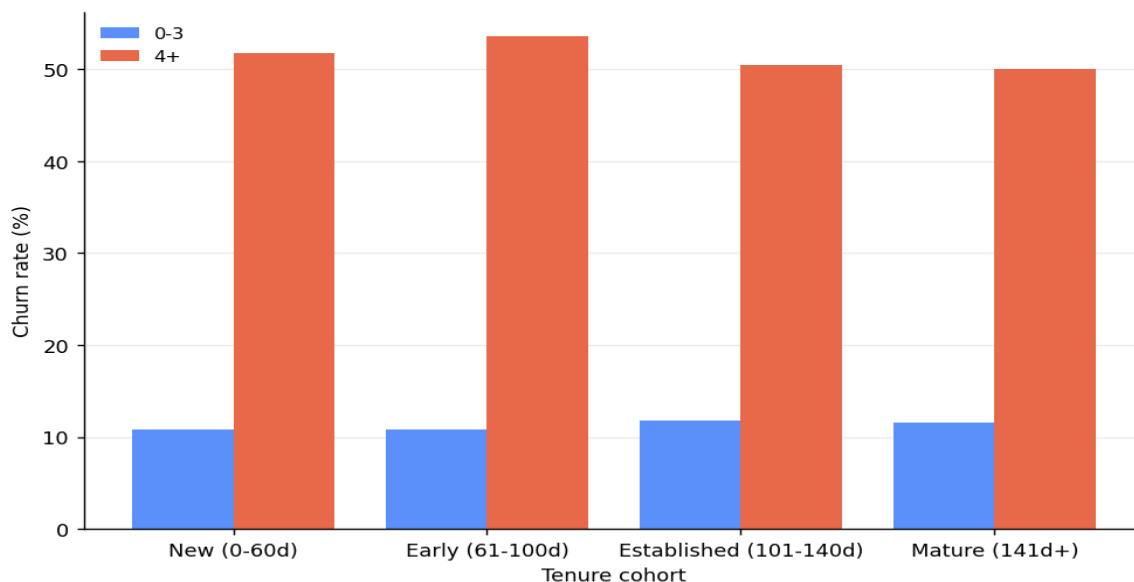


Plotted on the same y-scale as the driver charts to show how little variation there is here by comparison.

AI explanation. The chart shows churn rates that hover between 13.11 % and 15.29 % across the four tenure cohorts, with only a modest rise to 15.29 % in the Established group before slipping back to 14.12 % in the Mature group; the overall pattern is essentially flat rather than sharply changing at any point. This suggests that how long a customer has been with us does not meaningfully influence their likelihood to churn, ruling out tenure-based segmentation as a useful targeting lever. Because tenure is not a driver, the next step is to shift churn-prevention resources toward factors that do vary, such as customers with four or more service calls, and design interventions around that threshold.

Churn by tenure cohort and service-call bucket

The service-call cliff holds in every cohort. Tenure itself does not predict churn.



Tenure is the ordered axis in this project because the dataset has no date column. It is useful for structure, not as a churn driver.

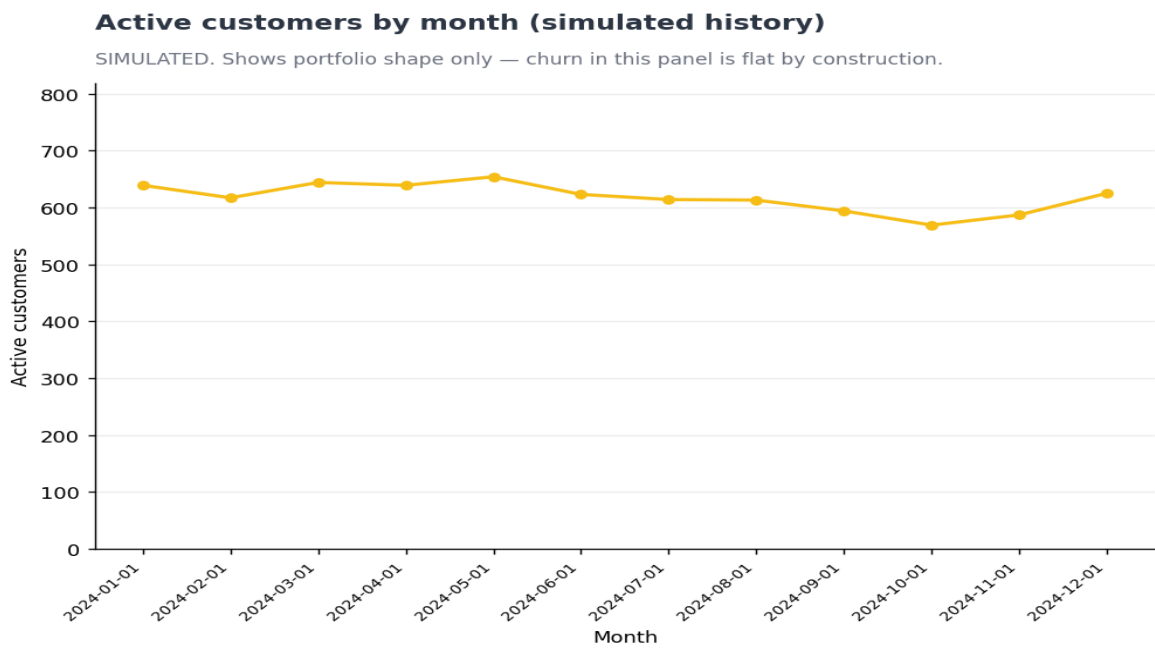
AI explanation. The churn rate stays near 10.79 %– 11.79 % for customers with 0–3 service calls across all four tenure cohorts, then jumps sharply to 50.0 %– 53.54 % for those with 4+ calls, and the jump is similar in each cohort. This pattern suggests that the volume of service calls is a structural driver of churn that outweighs any effect of how long a customer has been with us. To reduce churn, launch a proactive outreach program that contacts every customer as soon as they reach four service calls, resolves the outstanding issues, and offers a targeted retention incentive.

Portfolio structure over time‡

SIMULATED HISTORY. This section is derived from generated monthly snapshots, not observed history. It shows portfolio structure only. Churn in this panel is flat by construction and cannot support a trend claim or a forecast.

12	569-654	\$8,529-\$9,954
Months shown	Active base	Monthly revenue

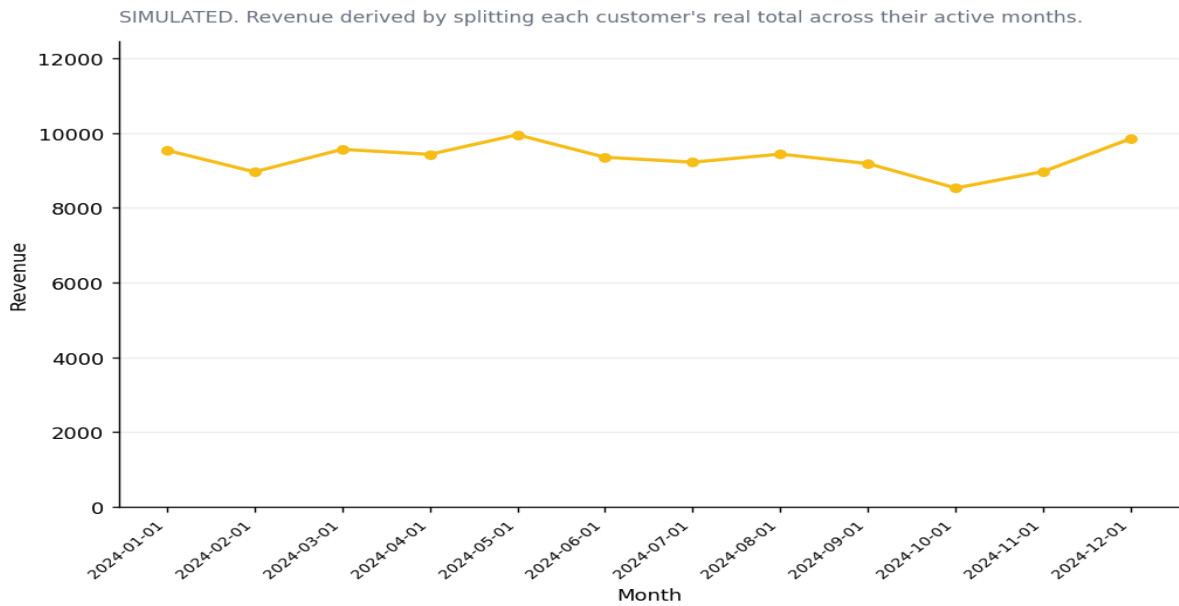
The simulated snapshot shows an active base that varies between 569 and 654 customers, generating period revenue between \$8,529.49 and \$9,954.10 with per customer charges (ARPU) ranging from \$14.53 to \$15.76. The portfolio is composed of customers exposed to three independent churn drivers – 4 or more service calls, an international plan subscription, and heavy daytime usage (day_charge ≥ 45) – which together define the risk profile of the base. Because churn is flat by construction, the primary lever for improving revenue is to reduce exposure to these drivers, for example by offering proactive support to the 4+ service call segment and bundled discounts to international plan users with high daytime usage. Implement a targeted retention program that identifies customers meeting any of the three risk criteria and delivers a tailored incentive to lower their likelihood of leaving.



Generated from the snapshot, not observed. Included to show portfolio shape; it contains no information the snapshot did not already hold.

AI explanation. The active customer line stays within a narrow band, hovering around six hundred plus, with the lowest point at 569 and the highest at 654, and a noticeable dip around month 10 before climbing back to 625. This shape suggests a simulated portfolio where entries and exits are roughly balanced, keeping the base stable. Continue the current acquisition and retention programs and align support staffing to the observed volume to preserve this balance. Remember that this history is simulated, so treat the numbers as a baseline for testing any new initiatives.

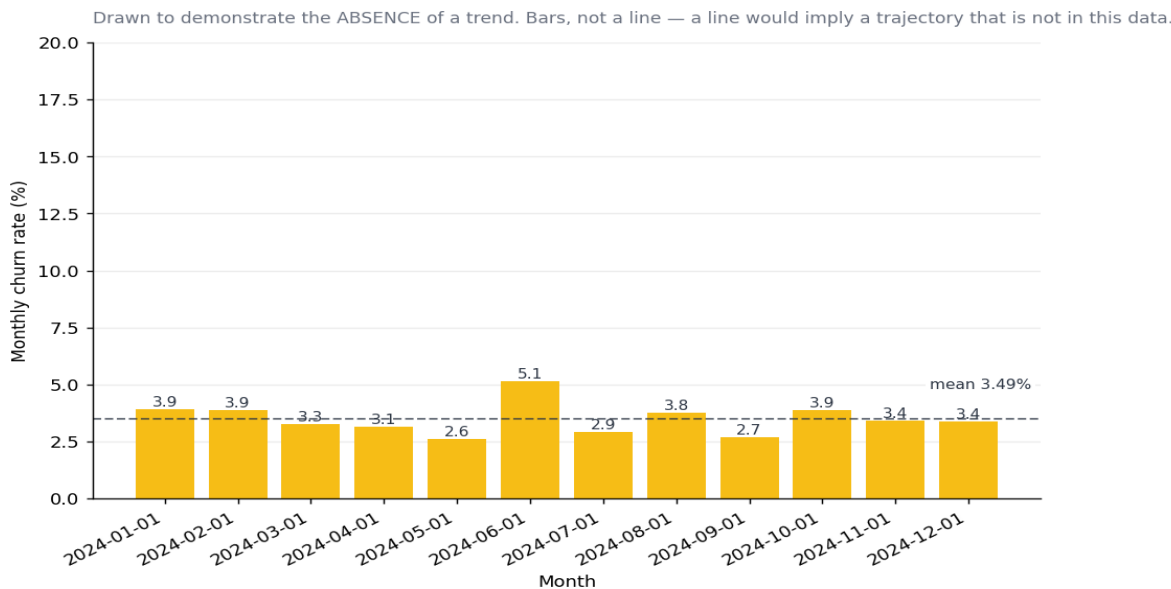
Monthly revenue (simulated history)



Each customer's real total charge was distributed across their active months. Monthly totals sum back to the real portfolio revenue.

AI explanation. The simulated monthly revenue hovers between roughly 9,500 and 9,600 for most months, with a noticeable dip to 8,529.49 in October before returning to the 9,800 range in December; the daytime share stays tightly clustered around 51 percent throughout. This pattern suggests the generator produces a generally stable revenue level but occasionally yields outlier months that are not driven by real customer activity. Because the variation is purely sampling noise, it does not reflect any underlying business shift. Adjust the simulation's random seed or variance parameters and rerun the model to confirm that the observed outliers remain within an acceptable tolerance.

Monthly churn is FLAT by construction (simulated)



Included to demonstrate the absence of a trend, not to show one. Churn timing in this panel was generated; a different random seed would produce a different arrangement of the same flat distribution. Real churn movement over time cannot be measured from a single snapshot.

AI explanation. The bars for each month sit tightly around the reference line of 3.49%, with individual churn rates ranging from about 2.6% to 5.14% and no abrupt jumps. This flat appearance tells us the simulated generator produced churn that is essentially constant, and the modest spread is just sampling noise rather than a business-driven signal. Because the chart's purpose is to show the absence of a trend, the concrete next step is to use this simulated series as a control when testing churn analysis tools, confirming that they do not flag false patterns.

What this report cannot tell you

- **NO TIME DIMENSION.** The source data is a single snapshot with no dates. Nothing here describes how any metric changed over time, and no figure in this report is a forecast.
- **SIMULATED HISTORY SHOWS STRUCTURE, NOT TRENDS.** The monthly panel in this report is generated from the snapshot, so it contains no information the snapshot did not already hold. Churn in it is deliberately flat: a trend there would reflect the generator's random seed, not customer behaviour. It is included to show portfolio shape and tenure lifecycle, both of which are legitimate uses of derived monthly data.
- **TENURE IS NOT A DRIVER.** Churn is flat across tenure cohorts (13.1%-15.3%). This was expected to matter and does not - a genuine finding, and a reason not to target by tenure.
- **STATE-LEVEL FIGURES ARE NOISY.** 51 states average about 65 customers each, so a handful of churners moves a state several percentage points. Geography is the weakest dimension here.
- **PROJECTED VALUES REST ON ASSUMPTIONS.** Save rate, contact cost and acquisition cost are industry-typical placeholders, not measurements: the dataset contains no cost data. A save rate of 30% is assumed, with a sensitivity band of 15%-45%.
- **REVENUE IS PERIOD CHARGES.** Not monthly recurring revenue and not annualised.
- **CHURN RISK IS NOT PREDICTED HERE.** Segments describe which customers resemble those who already churned. Scoring individual customers is a separate modelling problem.

Provenance

Every observed figure traces to the source data. Segment totals reconcile to the portfolio: customers match (3,333), revenue matches (\$198,146.03).

Warning: narration in these sections failed validation and may contain unverified figures: kpi.

Warning: the AI explanation beneath these charts failed validation and is marked in red: churn_by_plan.